# 1. What PumpPulse does

PumpPulse is a browser dashboard that helps you explore solar-powered irrigation pumps. You choose a well, select a pump design, change operating conditions and examine the estimated water, power and mechanical stress. Its purpose is to make the effect of each decision easier to understand.

## The problem it explores

A pump must deliver enough water, but running faster or under poor water conditions can increase stress. Sunlight also changes during the day, so the available electrical power is not constant. PumpPulse lets you compare these competing needs in one place.

## Dashboard

A dashboard is a screen, or collection of linked screens, that brings information and controls together. PumpPulse has sections for location, pump selection, engineering settings, replay, scheduling, comparisons and reporting.

## Simulation and digital twin

A simulation is a calculation-based imitation of a system. A digital twin is a digital representation linked to a particular system and its state. Here, the pump twin is an educational, simulated representation. It is not continuously connected to an installed physical pump.

## What the current version actually contains

The supplied enhanced demo runs in the browser using sample stations and synthetic hourly inputs. Pump designs and specifications are illustrative. The calculations are useful for learning and comparisons, but they have not been calibrated against a manufacturer pump curve or field measurements.

## Who can use it and why

Students can learn engineering concepts, project teams can explain their design choices, and reviewers can inspect how assumptions affect results. The application demonstrates a possible decision-support approach. Using it to control a real irrigation installation would require verified data and testing.

**How to read this guide:** Each section explains what a feature means, what it does and why it exists. Numerical examples are teaching examples unless explicitly identified as demo defaults. They are not measured project results.

# 2. Dashboard sections and reading path

## Start with the system

Mission Control gives the overview. India Atlas locates sample wells. Station Details records the selected well and replay time.

## Understand and configure the pump

Pump Model Explorer compares conceptual designs. The 3D Pump Twin reveals the assembly. Mechanical Parameters Lab changes the inputs. Engineering Performance explains the calculated outputs.

## Examine behaviour over time

Live Virtual Monitoring displays the current simulated hour. Forecast & Schedule explores the next 24-72 hours of synthetic conditions. Season Time Machine replays a selected demo day and compares alternative policies.

## Compare and explain results

Water-Life Ledger breaks down water and stress by hour. Policy Arena compares three operating rules. Trust Lab tests whether a recommendation changes when assumptions change.

## Look beyond one replay

Fleet Intelligence compares sample stations. Bearing Analysis demonstrates vibration-signal concepts. Evidence & Reports records settings and exports results.

## A sensible first experiment

Select Sample Well A, choose a submersible model and note the date and hour. Open the twin, then record water, power and wear. Increase the number of panels, apply the settings and compare the same policy at the same date. Change one input at a time so you can explain the difference.

## Selections and persistence

The selected well, model, policy and timestamp are shared across the dashboard screens. This keeps a comparison tied to the same scenario. Shared state during a browser session is different from permanent saving; the supplied build does not provide a user-account database or guaranteed session restoration.

# 3. Mission Control and status labels

### Mission Control

This is the home screen. It gives quick entry points into the map, twin, forecast and replay, plus summary figures. Its purpose is orientation: you can see the project focus before opening a detailed engineering screen.

### Stations loaded

A station is a location entry with a unique ID, name, state, district and sample well information. The demo contains six fictional stations. A count of six means six entries are available; it does not mean six live pumps are connected.

### Water delivered and wear budget

Water delivered is the simulated volume moved during the calculation period. Wear budget is a relative stress score used for comparison. A displayed percentage is based on a reference budget, not a medical-style health percentage for the machine.

### Selected well and scenario

The status bar identifies the active well and operating policy. A scenario is the combination of inputs used in an experiment: location, date, pump, configuration and uncertainty settings. Changing the scenario changes what is being compared.

### Mode: Live, Forecast or Replay

These names describe how information is presented. Live means stepping through a virtual operating state, Forecast means exploring future synthetic inputs, and Replay means revisiting a selected demo time. The names do not establish that real sensors or weather services are connected.

### Weather: demo and API: offline

These labels explain the source and connection state. An API is a way for one software system to request data or actions from another. Offline here means the dashboard uses local demo logic rather than calling the project backend.

### Disabled Open Twin button

A disabled button prevents starting the well-specific twin before a well is chosen. Station setup also checks the demo date range. This helps users avoid a screen with no meaningful location or timestamp.

# 4. India Atlas and well information

## Atlas, map boundary and state boundaries

An atlas is a map-based way to browse places. The national outline gives geographic context; thinner internal boundaries help identify states. The enhanced build embeds a legacy state-boundary dataset. Its depiction is a dataset convention, not a current legal statement about borders.

## Latitude and longitude

Latitude describes north-south position; longitude describes east-west position. They are measured in degrees. A station at 11 degrees north and 77 degrees east is placed using those coordinates. This is more meaningful than guessing a marker position on an image.

## Marker, selection ring and tooltip

A marker represents a well entry. Available demo wells are teal; the selected marker is brighter cyan with a ring. A tooltip is the small information box shown on hover. Clicking chooses the well and makes its details available elsewhere.

## Search, state filter and Selected only

Search matches station text such as name, district or ID. A state filter limits the list to one state. Selected only hides other stations. These controls reduce clutter; they do not improve the quality of the underlying data.

## Fit India, Outline and Show state boundaries

Outline controls boundary visibility. Show state boundaries controls internal lines. In this build, Fit India restores the boundary display in the fixed national view; it is not a full interactive map zoom-and-pan engine.

## Depth to water and observation date

Depth to water is the vertical distance from a reference ground level to the water surface, not the full depth of the borewell. The observation date says when a reading would have been taken. The displayed values and dates are fictional demonstration metadata.

## Coverage or completeness

Coverage normally means how much expected data is present. For example, 80% could mean 80 of 100 expected observations exist. In this demo, coverage percentages are sample labels; they are not computed quality audits of a real groundwater dataset.

# 5. Station setup, dates and time

## Station ID, state and district

An ID uniquely identifies a record even if two names are similar. State and district describe administrative location. These fields keep exports and comparisons tied to the intended sample station.

## Agency and provenance

Agency would identify the organisation responsible for measurements. Provenance means the origin and processing history of information. The supplied stations are explicitly fictional and should not be presented as Central Ground Water Board measurements.

## Date picker and hour slider

The date picker chooses a day; the hour slider chooses an integer hour from 0 to 23. The original demo period is 1 March to 31 May 2024. The hour is shown in IST, Indian Standard Time. It is a replay choice, not necessarily the current local time.

## Selected-hour values versus daily totals

A selected-hour value describes one hourly step. A daily total combines 24 steps. At 10:00, the side panel shows that hour; a daily water summary may still describe the full simulated day. Always check which time period a number represents.

## Pump selection and intake depth

Pump selection chooses the conceptual machine. Intake depth is how far below ground its water entrance is placed. The demo initially derives an intake estimate from the sample water depth plus assumed submergence. This is not an installed-pump survey.

## Open the Pump Twin

This action opens the virtual pump at the selected well and time. Its purpose is continuity: the station you choose on the map should be the station identified beside the pump and its graphs.

## What happens when you change wells or models

Changing wells updates sample lift and intake inputs. Selecting a new model restores that model's default configuration and clears the previous trust result. The well and replay timestamp remain selected. Review settings again because model switching is not a promise to preserve every custom value.

# 6. Pump types: what the names mean

## Submersible and multistage

Submersible means the pump is installed below water. Multistage means water passes through several pumping stages in sequence. Each stage contributes to the total pressure rise. The demo includes both a general multistage submersible and a higher-head deep-lift variant.

## Deep-lift and borewell

Deep-lift describes a design intended for a greater lifting requirement. A borewell is a narrow drilled well. The solar borewell multistage model is shown as a long vertical assembly. These labels overlap: a borewell pump can also be submersible and multistage.

## Surface centrifugal pump

Surface means mounted above ground. Centrifugal refers to the rotating impeller adding energy to water, followed by the casing converting part of that motion into pressure. A surface pump drawing through suction has a limited suction lift; it cannot simply pull water from any depth.

## Solar DC brushless

DC means direct current. Brushless means the motor uses electronic switching instead of mechanical brushes to control the rotating magnetic field. A controller manages electrical operation. This term describes motor technology, while submersible describes installation; they are not mutually exclusive categories.

## Helical rotor / progressive cavity

A helical rotor turns inside a matching stator and moves cavities of water along the assembly. The rotor is the rotating part; the stator is the surrounding stationary part. The name describes a different pumping mechanism from an impeller-based centrifugal pump.

## Surface irrigation system

This is a system-level arrangement: a surface pump, water source, delivery connection and solar array. It is not a completely separate hydraulic principle. The extra components help show the full irrigation setup.

## Why all seven can represent solar configurations

Solar power describes the energy source, not the pump's shape. A compatible motor, controller and sometimes an inverter connect the array to the pump. The seven entries are conceptual choices, not seven verified manufacturer products.

# 7. Model Explorer and specifications

## Preview, Select Model and Configure

Preview lets you inspect a model. Select Model makes it the active simulation model. Configure selects it and opens the parameter lab. The comparison selector places another model's specification values beside the active model.

## Model ID and manufacturer model number

A model ID is the application's identifier, such as DEMO-BLDC. A manufacturer model number identifies an actual commercial product. The demo does not supply verified manufacturer identities, so its internal IDs must not be presented as product references.

## Rated power, flow and head

Rated means a declared reference operating condition. Power is an energy-transfer rate. Flow is water volume per time. Head is the water-column equivalent of the energy needed to lift and pressurise water. A real rated flow and head should belong to a documented operating point.

## Maximum flow and maximum head

These are separate limits. A real centrifugal pump usually cannot deliver maximum flow and maximum head simultaneously. You need its performance curve to know which combinations are possible. Demo card limits are assumed and are not a validated curve.

## Nominal voltage, current and efficiency

Nominal means a reference value. Voltage is electrical potential difference; current is the flow of charge; efficiency is the useful output divided by input. The card's nominal current is a rough rating-based estimate, not a measured current trace.

## Supported operating modes

The surface centrifugal demo has OFF and ON at nominal speed. Other demo choices support stepped fractions 0, 0.4, 0.7 and 1. A selected LOW or MEDIUM state must correspond to a supported level. A dropdown label alone does not prove a physical controller supports that state.

## Hide incompatible designs

This filter evaluates each model using its demo defaults for the chosen well. It helps find obviously unsuitable combinations. It is not a certification search across real products, and a design that passes assumed limits remains conditional until verified data is available.

# 8. Inside the 3D Pump Twin

## Motor, shaft and coupling

The motor converts electrical energy into rotational mechanical energy. The shaft transmits rotation to the pumping mechanism. A coupling joins rotating parts, such as a surface motor shaft and pump shaft. Their purpose is to transfer torque: the turning effect of a force.

## Impeller, stage and casing

An impeller is a rotating bladed part that transfers energy to water. A stage is one pumping unit in a multistage assembly. The casing encloses and guides flow while providing structure. The conceptual viewer uses reusable shapes rather than exact manufacturing drawings.

## Bearing, inlet, outlet and controller

A bearing supports a rotating shaft and reduces friction between moving parts. The inlet or intake is where water enters; the outlet is where it leaves. The controller manages electrical operation. A highlighted bearing is not evidence that a real bearing fault has been detected.

## External, X-ray and exploded views

External view shows an assembled appearance. X-ray view makes housing transparent so you can see internal components; it is a rendering effect, not medical X-ray imaging. Exploded view separates parts for learning; it is not how a pump operates.

## Orbit, zoom and reset camera

Orbit rotates your viewing angle around the assembly. Zoom changes how large it appears. Reset restores the viewing setup. These change the camera view, not motor speed, water flow or engineering settings.

## Component inspection, labels and animation

Clicking supported parts shows a short explanation. Labels identify the assembly. Impeller motion, water markers and solar-panel glow follow the simulated state. Animation is a visual explanation, not a laboratory measurement of speed or fluid motion.

## Three.js, WebGL and conceptual geometry

Three.js is the JavaScript graphics library used for the scene. WebGL lets a browser use graphics hardware for 3D rendering. Geometry describes shapes and positions. If WebGL is unavailable, the build reports the limitation and may show its simpler canvas fallback. The design is not a computational-fluid-dynamics simulation.

# 9. Mechanical Lab: how controls work

## Parameter and configuration

A parameter is one input, such as pipe diameter. A configuration is the complete set of inputs. Changing one parameter can change several results, so the active configuration must be shared by the twin, graphs, ledger and comparisons.

## Bounds, validation and integer inputs

Bounds define allowed minimum and maximum values. Validation rejects empty, non-numeric or out-of-range entries. Counts such as panel count and stage count must be whole numbers. An input passing a numeric range still needs a separate compatibility check.

## Preview and debounce

A preview estimates the effect before committing settings. Debouncing waits briefly after typing before recalculating; the enhanced demo waits about 250 milliseconds. This prevents repeated calculations for every digit entered.

## Apply configuration and reset

Apply commits the draft values after validation and compatibility checks. It updates the shared calculation and daily delivery target and clears a stale trust result. Reset restores the selected model's baseline assumptions. Neither action physically changes a real pump.

## Explore mode

Explore mode lets you vary illustrative inputs and inspect their effects. Its purpose is experimentation: for example, compare a wider pipe with a narrower pipe while leaving every other setting unchanged.

## Verified configuration mode and override

Verified configuration mode locks baseline motor fields unless you explicitly enable an override. Override means replacing a baseline assumption for an experiment. The current models still have assumed specifications: the word Verified in the interface does not make them manufacturer-verified.

## Change summary and controlled comparison

The summary tells you what was previewed or applied and explains rejection. For a fair experiment, record a baseline, change one input, keep the same date and policy, apply and compare. Re-run Trust Lab after changing assumptions rather than quoting its old verdict.

# 10. Pump and motor parameters

## Rated motor power (W)

Watt is a unit of power. In the current demo this value is treated as a nominal electrical-input ceiling. A real motor datasheet may state rated shaft-output power instead, so its meaning must be checked before importing a specification. 1 kW equals 1,000 W.

## Voltage (V) and current (A)

Volt measures electrical potential difference; ampere measures charge flow. For a simple DC calculation, electrical power is voltage multiplied by current. The controller current setting acts as an input limit. Real AC power also depends on phase arrangement and power factor, which this demo does not model.

## Motor efficiency and hydraulic efficiency (%)

Motor efficiency is shaft output divided by electrical input. Hydraulic efficiency is water power divided by shaft input. With 85% and 60%, the combined useful-water fraction is 0.85 x 0.60 = 0.51, or 51%. The remaining energy goes into losses such as heat and friction.

## Nominal, minimum and maximum RPM

RPM means revolutions per minute. Nominal RPM is the reference shaft speed; actual demo speed equals the supported speed fraction multiplied by nominal RPM. Minimum RPM can switch off operation below the chosen threshold. Maximum RPM is checked for consistency with nominal RPM; it is not a detailed motor-protection model.

## Stages and impeller diameter (mm)

Stages represent successive pumping units. Impeller diameter is the size across the rotating component. In real machines both influence performance and must be matched to the design. Here they adjust a simplified flow bound; stage count also affects the conceptual submersible geometry, up to the viewer's visual cap.

## Shaft diameter (mm) and assembly mass (kg)

Shaft diameter describes its thickness; mass describes the assembly's amount of matter. The demo uses both in an assumed start-stress multiplier, and shaft diameter affects part of the geometry. This is not a calculated shaft-strength assessment, inertia model or bearing-life standard.

# 11. Groundwater, intake and head

## Static lift (m)

Static lift is the vertical lifting requirement before adding pumping-related water-level change or pipe loss. In this demo it is based on sample depth to water and assumes the relevant delivery reference is near ground level. Real elevated storage tanks or pressure requirements would add further head.

## Dynamic drawdown (m)

Drawdown is the fall in the water level caused by pumping. A deeper water surface increases lifting effort and reduces the water above a submerged intake. The input is a simplified assumed amount; the replay also uses an additional uncalibrated hourly water-level response for submergence.

## Intake depth and submergence (m)

Intake depth is the position of the water entrance below ground. Submergence is the water depth above that entrance. If intake is at 30 m and the pumping water level is at 22 m, submergence is 8 m. Intake depth is not added again to the lift calculation.

## Minimum submergence and dry-run risk

Minimum submergence is a chosen safety threshold. Too little water above the intake can create poor operating conditions. The demo flags dry-run risk when simulated submergence falls below the threshold while running; it does not prove that the intake is completely dry or detect actual cavitation.

## Total dynamic head (TDH)

TDH is the total energy requirement per unit weight of water, expressed as metres of head. The implemented breakdown is static lift + entered drawdown + pipe friction loss. A general field calculation can also need outlet elevation, pressure and fitting losses; those are not separate calibrated inputs here.

## A worked head example

Assume static lift is 20 m, drawdown is 3 m and pipe loss is 2 m. TDH is 25 m. If the intake is 32 m deep, the initial submergence is 32 - 20 - 3 = 9 m. The 32 m intake depth is not added to the 25 m head.

## Why deeper is not automatically more head

Installing an intake farther below an unchanged water surface increases submergence, not necessarily the water-surface-to-outlet elevation difference. This distinction prevents counting the same vertical distance twice.

# 12. Pipes, flow and water demand

## Pipe diameter (mm), length (m) and velocity

Pipe diameter is the inside-width assumption used by the calculation. Length is the flow-path distance. Velocity is how fast water moves through the pipe. For a fixed flow, a smaller diameter means higher velocity and often much greater friction loss.

## Friction factor and Darcy-Weisbach loss

The Darcy friction factor is a dimensionless coefficient representing pipe resistance. The demo uses a user-entered factor rather than calculating it from roughness and Reynolds number. Its formula is loss head = factor x (length / diameter) x velocity squared / (2 x gravity). Diameter must be in metres.

## Flow rate versus water volume

Flow is volume per time, such as 50 L/min. Volume is the total amount delivered, such as 3,000 L. At a constant 50 L/min for 60 minutes, water delivered is 3,000 L. The dashboard often displays L/h, which is L/min multiplied by 60.

## Target flow and required delivery

Target flow asks for a particular rate at an operating moment. Required delivery is a total daily goal in L/day. They are different: a pump can meet the hourly flow request but still miss the daily target because it runs for too few hours.

## Water density (kg/m³) and gravity (m/s²)

Density is mass per volume. Ordinary water is approximated as 1,000 kg/m³. Gravity, about 9.81 m/s², connects water mass to weight and lifting work. These inputs support hydraulic-power and pipe-loss calculations.

## Unit conversions

1,000 L = 1 m³. A flow in L/min is divided by 60,000 to obtain m³/s. A diameter in mm is divided by 1,000 to obtain metres. Unit conversion matters because a correct formula with mixed units can produce an answer thousands of times too large.

## What a pipe experiment teaches

Increase pipe diameter while keeping the same flow, lift and length. Calculated friction loss should fall. The saved energy requirement may allow more modelled delivery, but actual improvement still depends on the real pump curve and installation.

# 13. Solar and electrical parameters

## Panel rated capacity (Wp) and panel count

Wp means watt-peak: a panel rating under standard test conditions. An array is a group of panels. Ten 400 Wp panels give a 4,000 Wp array, but not a guaranteed 4,000 W at every hour. Series/parallel wiring and operating voltage are not explicitly modelled by panel count.

## Irradiance (W/m²)

Irradiance is sunlight power arriving per square metre. It is an input resource, not the pump's electrical power. The demo generates an hourly daylight pattern. Its irradiance control scales that pattern relative to 900 W/m²; it does not replace each hour with a constant measured value.

## Temperature derating (%)

Derating reduces rated capability under less favourable conditions. The demo's 90% temperature factor means retain 90% of the irradiance-adjusted array estimate. It is a simple retained-output factor, not a calculated panel-temperature model.

## Controller efficiency and wiring losses (%)

Controller efficiency is the fraction retained after controller conversion. A 95% value retains 0.95 of the incoming power. Wiring/conversion loss is the fraction removed: 5% loss retains 0.95. A higher efficiency increases output; a higher loss reduces it.

## DC direct drive and inverter-based AC drive

DC direct drive supplies a suitable DC motor through its controller. AC means alternating current; an inverter converts DC from the solar side into AC for a suitable motor. The architecture choice records the arrangement; the demo uses the entered loss factor rather than simulating inverter switching or AC waveforms.

## Why panel efficiency is not multiplied again

Panel efficiency normally compares incident solar energy with electrical output over panel area. When using a Wp rating already representing electrical output at a reference irradiance, multiplying another panel-efficiency factor would double-count conversion loss unless using a separate area-based model.

## Voltage compatibility

Matching an assumed nominal voltage is one demo check. Real compatibility also needs controller input range, panel string voltage, temperature limits and motor requirements. A sufficient array wattage alone does not establish electrical compatibility.

# 14. Following the energy calculation

## Step 1: estimate solar electrical power

The demo multiplies array Wp by hourly irradiance / 1,000, the irradiance scaling control / 900, the temperature factor, controller efficiency and retained wiring fraction. Trust experiments can apply another uncertainty multiplier. This is a simplified available-power estimate.

## Solar worked example

Take 10 panels rated 400 Wp each, hourly sunlight 800 W/m², default irradiance scaling, 90% temperature retention, 95% controller efficiency and 5% wiring loss. Available power is 4,000 x 0.8 x 0.90 x 0.95 x 0.95 = **2,599.2 W**. The example separates the 4,000 Wp rating from actual estimated output.

## Step 2: apply electrical limits

The demo limits pump input by available solar power, its speed-dependent rated-power estimate, and voltage x controller-current limit. A 2,200 W nominal pump cannot consume more than 2,200 W at its full-speed setting in this model, even if more solar power is available.

## Step 3: account for motor and hydraulic efficiency

If electrical input is 2,000 W, motor efficiency is 85% and hydraulic efficiency is 60%, the useful hydraulic budget is 2,000 x 0.85 x 0.60 = **1,020 W**. This budget bounds the power actually transferred to the water.

## Step 4: relate flow, head and hydraulic power

Hydraulic power = density x gravity x flow x head. Flow must be in m³/s. For 60 L/min, flow is 0.001 m³/s. At 25 m head, power is 1,000 x 9.81 x 0.001 x 25 = **245.25 W**. With the same efficiencies, idealised electrical input for that water power is 245.25 / 0.51 = about **481 W**.

## Power, energy and unused capacity

Power is the rate at one moment; energy is power accumulated over time. A constant 2 kW for 3 hours uses 6 kWh. Extra array capacity is not automatically consumed by the pump. The demo has no battery-storage or grid-export simulation.

**Important:** Enough energy is necessary but not sufficient for a real operating point. A physical pump must also satisfy its measured pressure-flow behaviour.

# 15. Engineering Performance and graphs

## Available solar, input power and hydraulic power

Available solar is the estimated electrical resource. Input power is what the simulated pump uses. Hydraulic power is the useful rate of energy delivered to water. Hydraulic power must stay below electrical input after losses. Zero flow may mean the model cannot support useful operation at that hour.

## Operating point and performance curve

An operating point is a flow/head combination. A real pump curve plots performance, often head against flow at a defined speed. A system curve describes the installation's required head at each flow. Their intersection determines operation. The dashboard has no validated curve, so its point is only a simplified estimate.

## How the demo finds its point

The calculation searches for a flow that fits its energy budget, assumed flow bounds and speed-scaled head limit. It uses a numerical binary search, repeatedly narrowing a candidate interval. This is a calculation technique, not proof that the chosen point lies on a real manufacturer curve.

## Sunlight versus speed graph

The horizontal axis shows hour. Sunlight and speed are separate series. The shared time cursor marks the inspected hour. Each series is scaled against its own maximum in the current chart helper, so equal visual height does not mean equal numeric values. Read the legend units and maxima.

## Cumulative water versus wear graph

Cumulative means a running sum. Each water point includes earlier hourly delivery; each wear point includes earlier stress scores. The lines show how both build up. Their vertical positions use different units and independent scaling, so a visual crossing is not a scientific threshold.

## Hourly and cumulative water or wear

Hourly describes one time step; cumulative adds time steps through the selected hour. For example, 2,000 L in hour 1 plus 3,000 L in hour 2 gives 5,000 L cumulative. The same distinction applies to relative wear.

## Water per wear unit

This ratio is delivered litres divided by relative wear. Higher can mean more delivery per modelled stress unit, but always check total target delivery too. If wear is zero, the ratio is undefined and the interface should show a dash rather than infinite efficiency.

# 16. Live monitoring and replay controls

## Live Virtual Monitoring

This screen presents the simulated current hour as if it were a monitoring view. It includes irradiance, solar output, motor speed, flow, input power, submergence, wear rate and operating mode. It is a virtual demonstration, not an Internet of Things sensor connection.

## RUN, OFF, warning and risk

RUN means the simulated pump has positive operating speed; OFF means it is stopped. A low-submergence warning highlights approach to the configured threshold. A start event records an OFF-to-running transition. A risk label comes from those assumed rules, not an observed fault diagnosis.

## Play, pause, restart and playback speed

Play advances replay time; pause freezes the selected time; restart resets to the first hour. Multipliers such as 5x or 20x accelerate the demo clock relative to its base playback rate. They do not multiply actual pump RPM or claim the animation is a real-time motor recording.

## Hour slider, graph click and event jump

The hour slider moves directly to an hourly state. Clicking a supported chart selects a time. Event buttons jump to starts or low-water-risk events. Their purpose is investigation: find why a wear increase occurred instead of watching the entire sequence.

## Season Time Machine

The page name describes a historical exploration idea. The supplied version lets you select a day in the demo period and inspect 24 hourly steps. It does not integrate and validate a complete multi-month pump season. Each selected day is recalculated using synthetic inputs.

## Reconstructed level and observation anchors

Reconstruction means estimating missing values between available information. An anchor is a point used to guide that reconstruction. The replay's displayed anchor dots and reconstructed label are demonstration visuals, not a proven reconstruction from actual well measurements.

## Branch Timeline and fair comparison

A branch shows an alternative policy under the same selected inputs. This supports a what-if question: what if another rule had operated the pump? It does not mean two different real histories occurred. Keep weather, well and uncertainty scenario consistent between branches.

# 17. Forecasts and operating policies

## Forecast horizon

A horizon is how far ahead a forecast extends. The dashboard offers 24, 48 or 72 hours. It generates synthetic variations for successive days. These are scenario inputs, not a connected meteorological forecast or a groundwater prediction validated in the field.

## Water target, shortfall and target met

The water target is desired delivery. Shortfall is max(0, target - delivered). Target met is delivered / target x 100%. If the target is 36,000 L and delivery is 30,000 L, shortfall is 6,000 L and target met is about 83.3%. This is a delivery metric, not model accuracy.

## Policy P0: fixed schedule

A policy is a rule for deciding how to operate. P0 requests full operation during the demo window, 08:00 through 17:00. Solar limits and overall configuration checks still constrain execution. Fixed schedule is the simple comparison baseline, not an unrestricted command overriding all limits.

## Policy P1: threshold protection

P1 requests full operation in the same window only when submergence exceeds the configured minimum. A threshold is the cutoff used in that decision. It illustrates a simple protective rule rather than a learned controller.

## Policy P2: wear-aware scheduler

P2 requests 70% speed when submergence is comfortably above the minimum, 40% when closer to the limit, and OFF below it. The actual command is also restricted by solar power and supported model levels. This is a stepped heuristic, not a mathematically optimised schedule or trained AI.

## Supported states can change the outcome

The fixed-speed surface model only supports 0 and 1. A requested fraction such as 0.7 is rounded down to a supported value by the demo, which can leave that model OFF under P2. This reflects the current rule implementation, not a universal conclusion that surface pumps cannot be scheduled.

## Schedule blocks and risk events

Hourly blocks show the resulting OFF/LOW/MEDIUM/HIGH pattern. Risk events count model flags during the horizon. Compare the pattern with available power and delivery shortfall; the presence of coloured blocks alone does not show the best plan.

# 18. Water-Life Ledger and wear terms

## Ledger and modelled life budget

A ledger is an organised record. This one records hourly water, energy, stress and causes. Its purpose is accountability: you can trace a daily total back to individual hours. Life budget is a reference-score metaphor, not measured material loss or remaining service life.

## Speed-stress contribution

The demo uses a speed-fraction-cubed term multiplied by an assumed coefficient. At fraction 0.7, the cubic factor is 0.343; at 1.0 it is 1.0. This makes higher speed more costly in the chosen stress model. It is not a universal fatigue law for every pump technology.

## Low-submergence or dry-run contribution

The extra penalty applies when the pump runs below the selected submergence limit. It illustrates an undesirable condition. A real dry-run detector would need suitable measurements and validated thresholds; this simulated flag does not establish a physical failure mechanism.

## Starts, ramp severity and startup frequency

A start is counted when the previous hourly speed is zero and the new speed is positive. Ramp severity is an assumed multiplier on the start contribution. Startup frequency is derived from these transitions; it is not an independent frequency input. The code does not resolve fast within-hour starts or separately calculate ramp-down stress.

## Coefficients and sensitivity range

A coefficient sets how much one contribution counts. The demo startup, speed and dry-run coefficients vary within plus or minus 50% of assumed baselines 0.6, 0.9 and 2.5. A sensitivity experiment changes them to see whether conclusions depend heavily on those choices.

## Hourly wear, cumulative wear and main cause

Hourly wear adds that hour's contributions. Cumulative wear adds all hours so far. Main cause identifies the largest contribution in an hour. If total relative damage is 8 units, a budget display of 8% is a reference convention; it does not mean the pump has exactly 92% of its physical life left.

## Run-hours, energy and row inspection

Run-hours count hourly steps with positive speed. Energy sums hourly power over one-hour intervals. Clicking a ledger row returns to the corresponding hour in the twin. High-speed duration and start counts are coarse hourly proxies; there is no calibrated bearing-life calculation attached to this ledger.

# 19. Policy Arena and Trust Lab

## Baseline, feasibility and winner

A baseline is a simple reference for comparison. Feasible means satisfying the experiment's chosen requirement. The arena treats a policy as delivery-feasible if it supplies at least 80% of the daily target, then favours the smallest wear score among those candidates. If none qualify, it declares no winner.

## Why both water and wear matter

A policy that never runs can have very low wear but deliver no useful water. The feasibility requirement prevents treating this as the best irrigation plan. Conversely, a high-delivery plan may have more stress. The dashboard makes that tradeoff visible rather than hiding it in one number.

## Uncertainty, scenario and sensitivity

Uncertainty means an input is not precisely known. A scenario is one possible set of values. Sensitivity is how much results change when inputs change. Trust Lab perturbs wear, water-level shift and solar-conversion assumptions, then repeats policy comparisons.

## Scenario count, seed and reproducibility

Scenario count is the number of repeated trials, typically selectable from 10 to 300. A seed starts the pseudo-random number sequence. Reusing the same seed and inputs produces the same sequence, making an experiment reproducible. This is not a measure of random sampling quality in the real world.

## Win frequency, feasibility frequency and verdict

Win frequency is trials won / trials tested. Feasibility frequency is trials meeting the target / trials tested. The demo labels a leading policy RECOMMENDED at 80% or more wins, CONDITIONAL at 50-79%, and INSUFFICIENT EVIDENCE below 50% or with fewer than 30 trials. These thresholds are project choices, not statistical confidence levels.

## Worst damage, worst shortfall and regret

Worst means the largest value among tested scenarios, not every physically possible future. Regret compares a feasible policy's damage with the winning feasible damage in the same scenario. The displayed mean divides accumulated regret by all trials, so read it alongside feasibility frequency.

## Ranking reversal and limits of trust

A reversal occurs when a scenario changes the winner from the baseline case. Its explanation identifies one such case. A stable result means stable within tested assumptions; it does not prove a real-world optimum, safety certificate or failure probability. Trust Lab is an evaluation method, not a fourth pump controller.

# 20. Fleet and bearing-signal analysis

## Fleet Intelligence

A fleet is a group of devices or stations. The table brings sample location, coverage, model ID, simulated delivery, wear, preferred policy and trust verdict together. Its purpose is scanning several records quickly. The current demo reuses the active configuration across sample stations; it does not contain separate verified installed-pump inventories.

## Bearing fault, vibration and waveform

A bearing fault is damage or abnormal operation in a rotating support. Vibration is motion that oscillates with time. A waveform shows signal amplitude over time. Bearing Analysis generates healthy-like and faulty-like synthetic signals to teach feature extraction; it is not processing real farm-pump recordings.

## Sampling rate and frequency

Sampling rate is how many measurements are taken per second, expressed in samples/s or Hz. Frequency is how often a signal pattern repeats per second, also in Hz. The demo creates 256 values at a nominal 2,048 samples/s, a short illustrative signal window.

## Spectrum and Fourier transform

A spectrum separates a signal into frequency components. A Fourier transform is the calculation used to do this. The demo uses a simple discrete Fourier transform to obtain magnitudes. Its generic chart helper retains hour-style ticks, so those tick labels are not calibrated vibration time or frequency axes.

## RMS and peak

RMS means root mean square: square the samples, average them, then take the square root. It summarises signal magnitude. Peak is the largest absolute sample. Both depend on units and sensor scaling; the demo uses arbitrary units, shown as a.u., rather than measured acceleration.

## Crest factor and kurtosis

Crest factor is peak / RMS, indicating how large a peak is relative to typical magnitude. Kurtosis describes concentration in distribution tails and sensitivity to extreme values; the code uses the fourth central moment divided by variance squared. Neither feature alone proves a specific bearing defect.

## Feature extraction, classifier and HUST

Feature extraction turns raw samples into useful summaries. A classifier assigns categories using rules or a trained model. No trained classifier or real HUST recordings are bundled. Also, the displayed waveform uses absolute sample values, so it is a magnitude-like view rather than the complete signed signal.

# 21. Evidence, labels and software terms

## Evidence & Reports

This screen records the selected station, pump, policy, date, assumptions, seed and version. Its purpose is traceability: another person should understand what produced the results. An export records an experiment; it is not proof that the underlying assumptions are correct.

## JSON, CSV and Copy JSON

JSON is a structured text format that can hold nested settings and hourly results. CSV is a plain-text table whose columns are separated by commas. The demo JSON includes configuration, original model assumptions and calculated overrides. The CSV exports selected hourly result columns. Copy JSON puts text on the clipboard.

## Experiment ID, timestamp and version

An experiment ID labels a run. The demo generates a station-based label, so it should not be treated as a globally unique database ID. A timestamp records export creation time. A version identifies which application definition generated it; version changes may affect comparisons.

## OBSERVED and RECONSTRUCTED

OBSERVED should mean directly recorded information, with source and timing. RECONSTRUCTED should mean estimated from incomplete records. The app supports these labels, but the presence of a label does not convert fictional anchors or simulated values into actual observations.

## MODELLED, ASSUMED, DEMO and SYNTHETIC

MODELLED means calculated using a model. ASSUMED means chosen without verified evidence for this installation. DEMO means used for demonstration. SYNTHETIC means artificially generated. A result can be modelled from synthetic inputs and depend on assumed parameters at the same time.

## FORECAST, NASA POWER and CGWB

FORECAST means an estimate of future conditions; this build uses synthetic scenarios. NASA POWER is a potential external source of satellite/model-derived environmental data, not a local well sensor. CGWB refers to India's Central Ground Water Board; the supplied station entries are not CGWB measurements.

## Frontend, backend, shared state and offline

Frontend is the interface running in the browser. Backend is a server providing services, storage or calculations. Shared state is the currently selected information used by multiple screens. Offline demo means the provided local simulation can run without live service calls. This build does not provide login, accounts, real-device control or a connected backend.

# 22. Putting it together: an example

## The question

Suppose your project asks: can this configuration deliver 36,000 L in one day while keeping the modelled wear score low? Start with a sample well, one pump model, one date and one daily target. Record all four so the comparison can be repeated.

## First, check the installation

Assume lift is 20 m, drawdown is 3 m and intake is 32 m. Initial submergence is 9 m. If the minimum is 5 m, the starting intake condition clears that chosen threshold. Calculate pipe loss at the target flow and compare total head against assumed model limits. A surface suction model may fail even if its maximum head looks adequate.

## Next, check energy and compatibility

Assume the solar example from page 14 gives about 2,599 W. This is available resource, not a command to use all of it. Check motor rating, nominal voltage and controller current limit. The demo may reject a configuration at the chosen target-flow requirement even if some lower-flow operation could be physically possible; its checks are simplified design rules.

## Then compare the operating rules

Run P0, P1 and P2 with the same inputs. If an illustrative result delivers 30,000 L, it reaches 83.3% of the 36,000 L target and leaves 6,000 L shortfall. It passes the arena's 80% experiment threshold but does not meet 100% of the stated demand. Explain both facts.

## Inspect why the difference occurred

Look at supported speed levels, solar availability, submergence and the ledger. A low-wear policy may operate fewer hours or at lower speed. Click a high-stress hour to inspect the cause. The graph indicates a pattern; the hourly table supplies the specific values.

## Test the conclusion and export

Increase uncertainty ranges and rerun Trust Lab. If rankings reverse, explain which changed assumption caused the reversal. Export JSON to preserve settings and CSV for the hourly table. In your conclusion, say the result applies to this illustrative model and tested scenarios, rather than claiming a verified pump-life extension.

# 23. Common questions and interpretation

### Why is the pump stopped when there is sunlight?

Sunlight may still be insufficient for a supported speed level. The model may be outside the permitted time window, below its minimum RPM, below the chosen submergence limit, unable to satisfy a head bound, or marked incompatible. Read the compatibility explanation before assuming the viewer is broken.

### Why did adding panels not increase water?

The pump may already be at its supported speed, electrical-input ceiling or flow bound. Another limit can dominate, such as required head or the policy command. More potential solar energy does not guarantee more useful pumping.

### Does a more powerful pump always deliver more?

No. The configuration also needs compatible voltage, current capacity, water availability, flow/head performance and a suitable controller. The dashboard's assumed limits and stepped policies may prevent a higher-rated model from using its full capacity.

### Does a low wear score mean a healthy pump?

No. It means a low value under the chosen relative-stress formula. A real pump could have corrosion, debris, existing bearing damage, poor installation or other problems that the demo never models.

### Is P2 an AI algorithm?

The supplied P2 logic is a hand-written rule using thresholds and discrete speed requests. There is no trained machine-learning controller shown. Trust Lab performs repeated uncertainty experiments; repetition and randomness alone do not make the controller learned AI.

### Does CONDITIONAL mean safe to install?

No. In this build it means the assumed limits pass with unresolved verification. Real installation needs a measured well profile, manufacturer pump curve, controller/array design, electrical protection and appropriate engineering review.

### What is the strongest claim the demo supports?

It can explain how selected assumptions and operating rules change calculated water, energy and relative stress in a repeatable educational experiment. It cannot establish actual failure dates, measured water savings or field-validated control performance without additional evidence.

# 24. Quick reference and project purpose

## Units at a glance

W = watt, power. kW = 1,000 W. Wp = panel power rating at reference test conditions. kWh = energy. V = volt. A = ampere. RPM = revolutions per minute. L/min and L/h = flow. L/day = daily volume goal. m = metre. mm = millimetre. kg = kilogram. Hz = cycles per second. u = an arbitrary relative-wear unit.

## Water concepts at a glance

Lift = elevation requirement. Drawdown = pumping-related fall in water level. Intake depth = entrance position below ground. Submergence = water above the intake. TDH = total lifting/pressure/loss requirement expressed as head. Flow = rate; cumulative delivery = running total.

## Decision concepts at a glance

Policy = operating rule. Threshold = decision cutoff. Baseline = reference approach. Feasible = satisfies an explicitly chosen requirement. Sensitivity = response to an input change. Uncertainty = imperfect knowledge. Reproducible = repeatable with the same defined inputs and calculation.

## What makes the dashboard useful

It connects a map, conceptual machine, engineering controls, time-based outputs and decision explanations. That connection helps users ask why a result changed, compare alternatives fairly and communicate assumptions instead of presenting unexplained numbers.

## What is needed for real-world conclusions

Measured water levels and delivery, actual installation geometry, verified motor/controller/panel specifications, validated pump curves and field calibration would be needed. Long-term wear or failure claims additionally need appropriate condition data and independent validation. These are data and validation needs, not features already connected in this demo.

## Scope of this report

This guide describes the enhanced PumpPulse.html demo, identified in its exports as frontend-demo-0.2. It explains the supplied interface and calculation rules. It does not claim to be a browser test report or a manufacturer manual. No new dashboard features were added while writing this guide.

**Source basis:** Supplied enhanced PumpPulse.html, including navigation, parameter definitions, simulation, compatibility checks, trust experiments and export logic. Map attribution in the application: geohacker/india legacy state GeoJSON (github.com/geohacker/india). Teaching examples in this report are hypothetical and explicitly separated from measured data.